



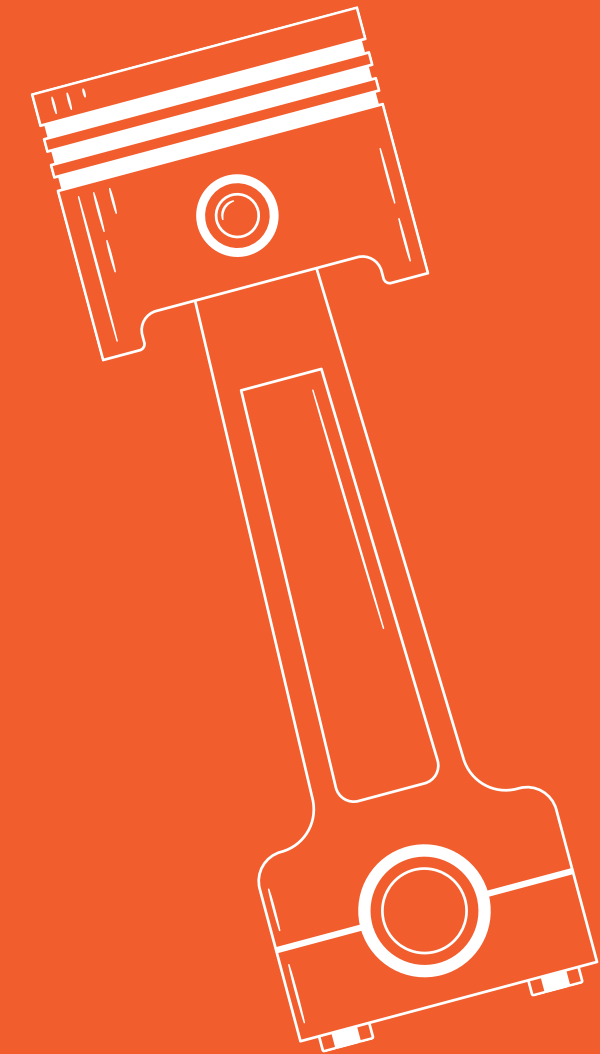
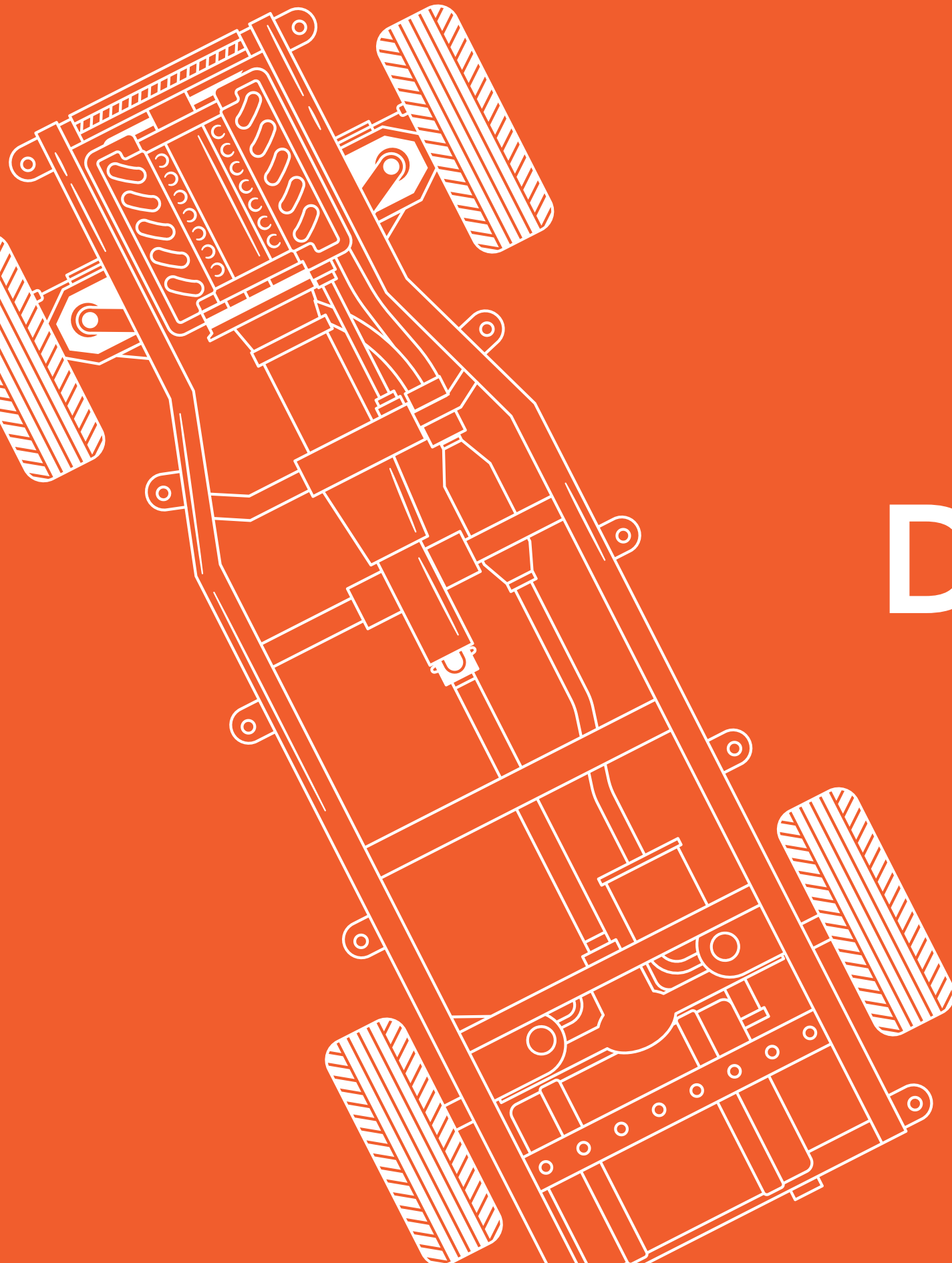
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TEAM B – MINI 005



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Competition Strategy



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Competition Strategy

Measure



Validate



Improve



Compete





Competition Strategy: Task Organization

Yellow: to be
done

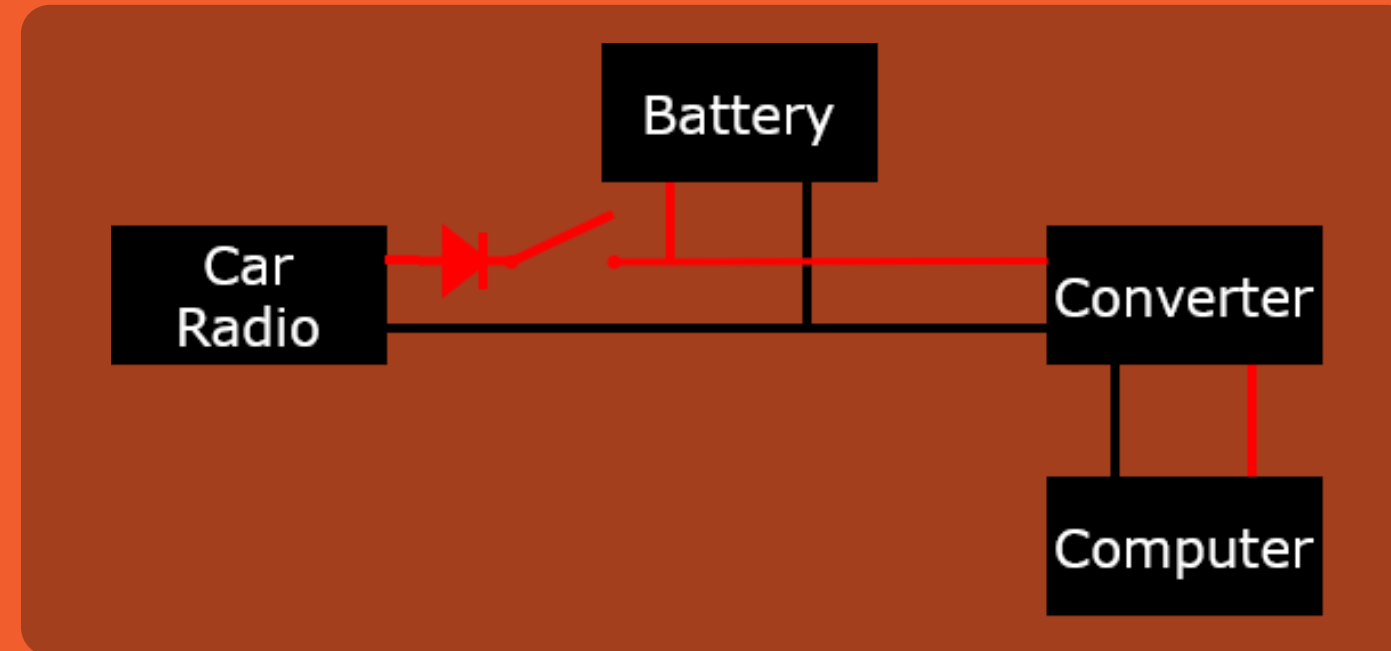
Green: done

Section	Tasks	Person	Deadline
Setup Challenge	Solder and Assemble Power Supply	Jon, Jan	04/08 - 11:00:00
	Measure cable lengths for computer and monitor	Jon	04/08 - 11:00:00
	unbolt and move IMU and fasten in front of car in/ near cup holder	Henry	04/08 - 11:00:00
	Move antenna to directly above IMU, re route, remove duct tape from on top	Henry, Aljaž	04/08 - 11:00:00
	Find connection points for ground	Hans	04/08 - 11:00:00
	Cable Manage	Jan	04/08 - 12:00:00
	Calibrate IMU, (read up the documentation of how)	Arthur, Martina, Dilan	04/08 - 11:00:00
CAN Challenge	Decode	Hans	
Sound Managment Challenge	car modifications	Aljaž	
	Tuning filters	Matteo,	
Precision Driving Challenge			
Misc	Continue Working On set up, add number to driver screen	Matteo	
	Look at shop, anything we need to buy?	Arthur, Martina, Dilan	
	Clean car, cable tire broken bits, add on stickers	Arthur, Martina, Dilan	
	Test powering the DAQ from PC	Henry, Jon	
	Start reading up on finance challenge, make a start	Arthur, Martina, Dilan	
	PPT slide on strain gauge	Aljaz, Jon	
	PPT slide on Marketing	Martina, Dilan	
	PPT slide on Special Challenge	Henry	
	Finish creating Content for video	Martina	
	Edit Video	Martina, Dilan	
	Prepare Data and files for submission	Matteo	
	Analyse oversteer data	Everyone	

PASS MOUNTING & SETUP IMMEDIATELY

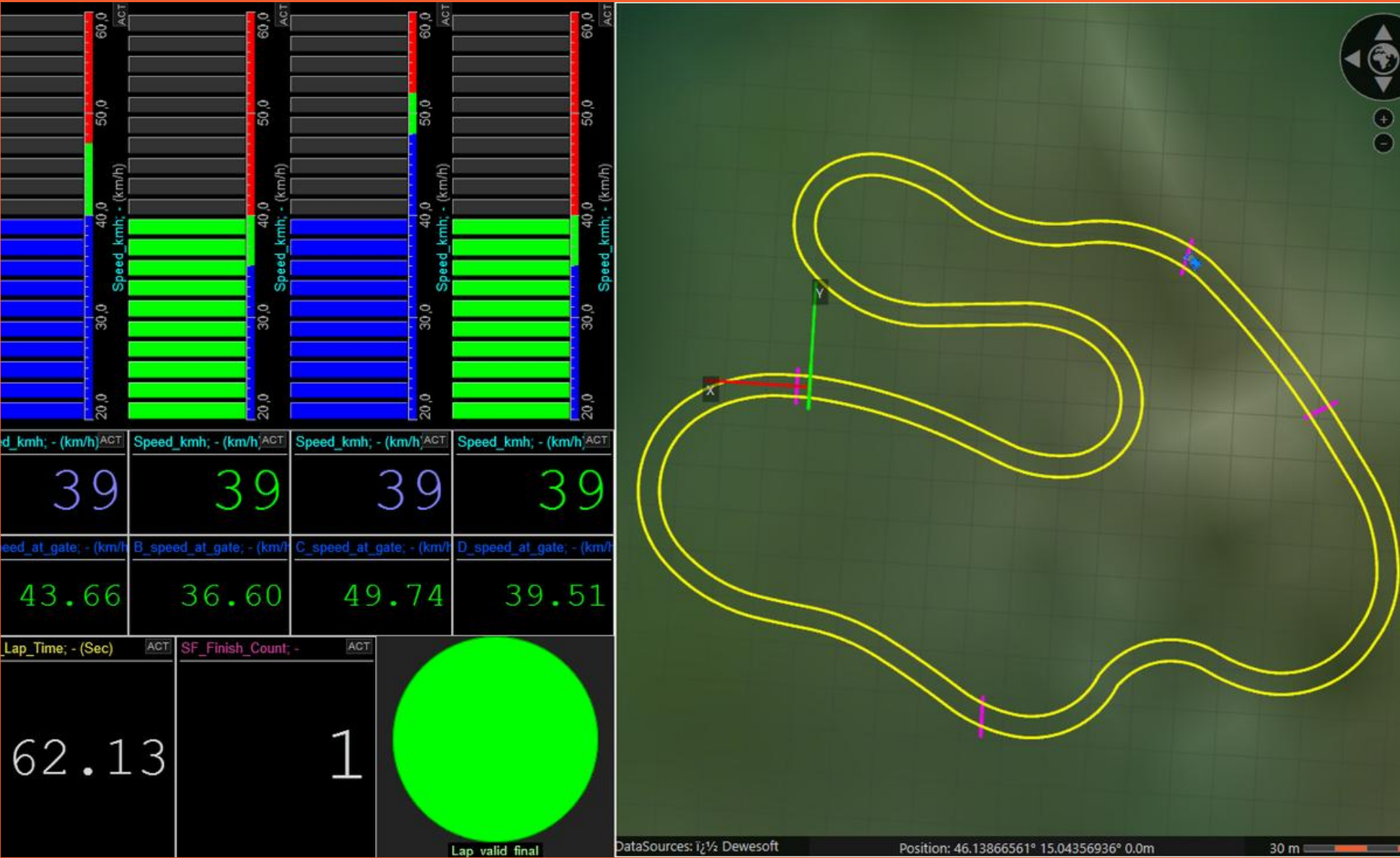
SOLUTION – PROGRESS – DIFFICULTIES – LEARNINGS

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OPTIMISE PRECISION DRIVING

SOLUTION – PROGRESS – DIFFICULTIES – LEARNINGS



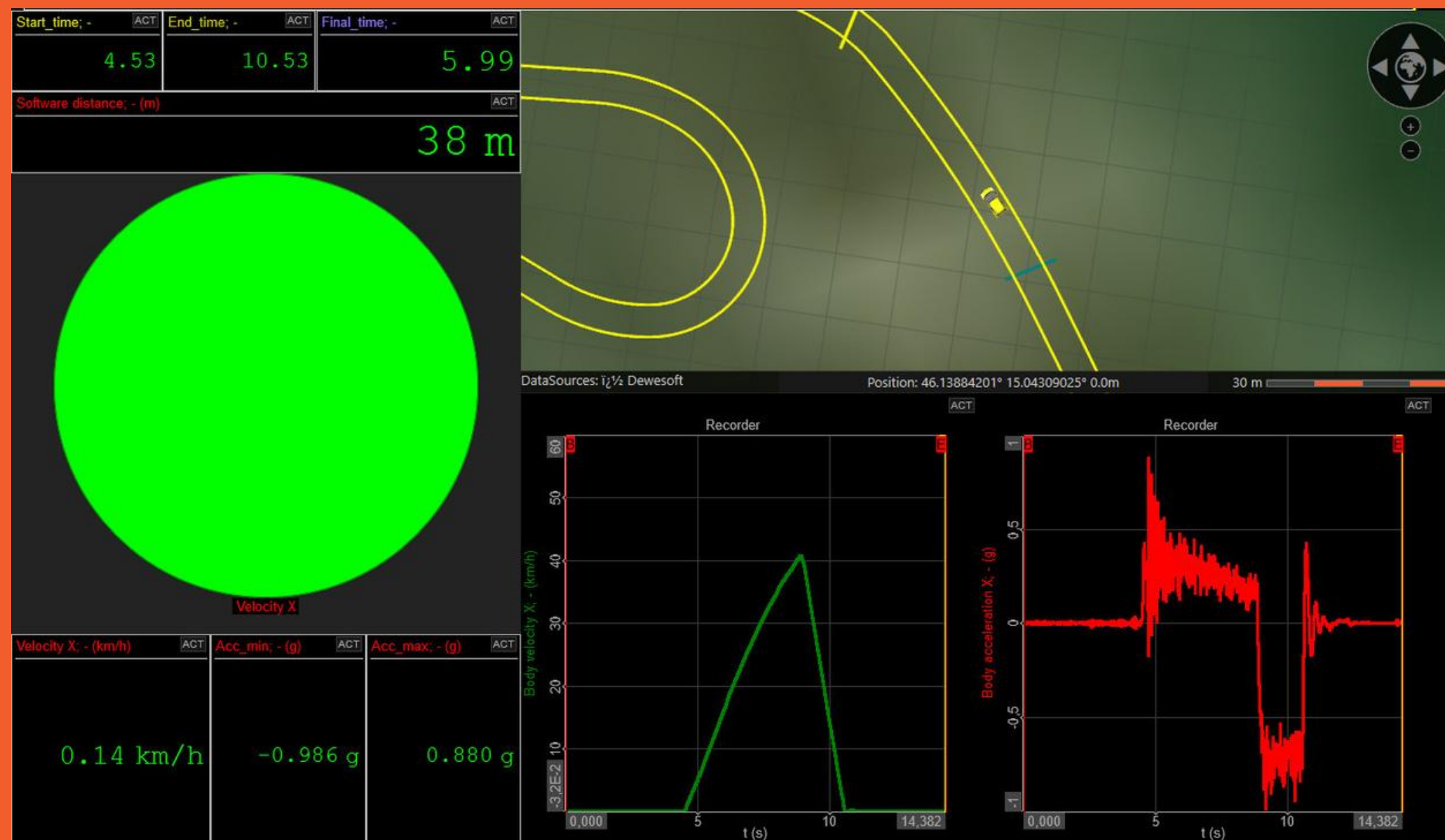
- Total valid laps: 6
- Challenge: assembling the setup screen



Run	Saving Laptop	Filename	Driver	Co-driver	Valid/Invalid	LapTime (only for valid) [s]
9	Matteo	PrecisionDriving_Challenge_0009	Henry	Matteo	1	62.13.00
10	Matteo	PrecisionDriving_Challenge_0010	Henry	Matteo	1	64.27.00
18	Matteo	PrecisionDriving_Challenge_0018	Jan	Matteo	1	61.16.00
33	Matteo	PrecisionDriving_Challenge_0033	Matteo	Aljaž	1	62.13.00
34	Matteo	PrecisionDriving_Challenge_0034	Matteo	Aljaž	1	62.24.00
38	Matteo	PrecisionDriving_Challenge_0038	Jon	Matteo	1	62.29.00

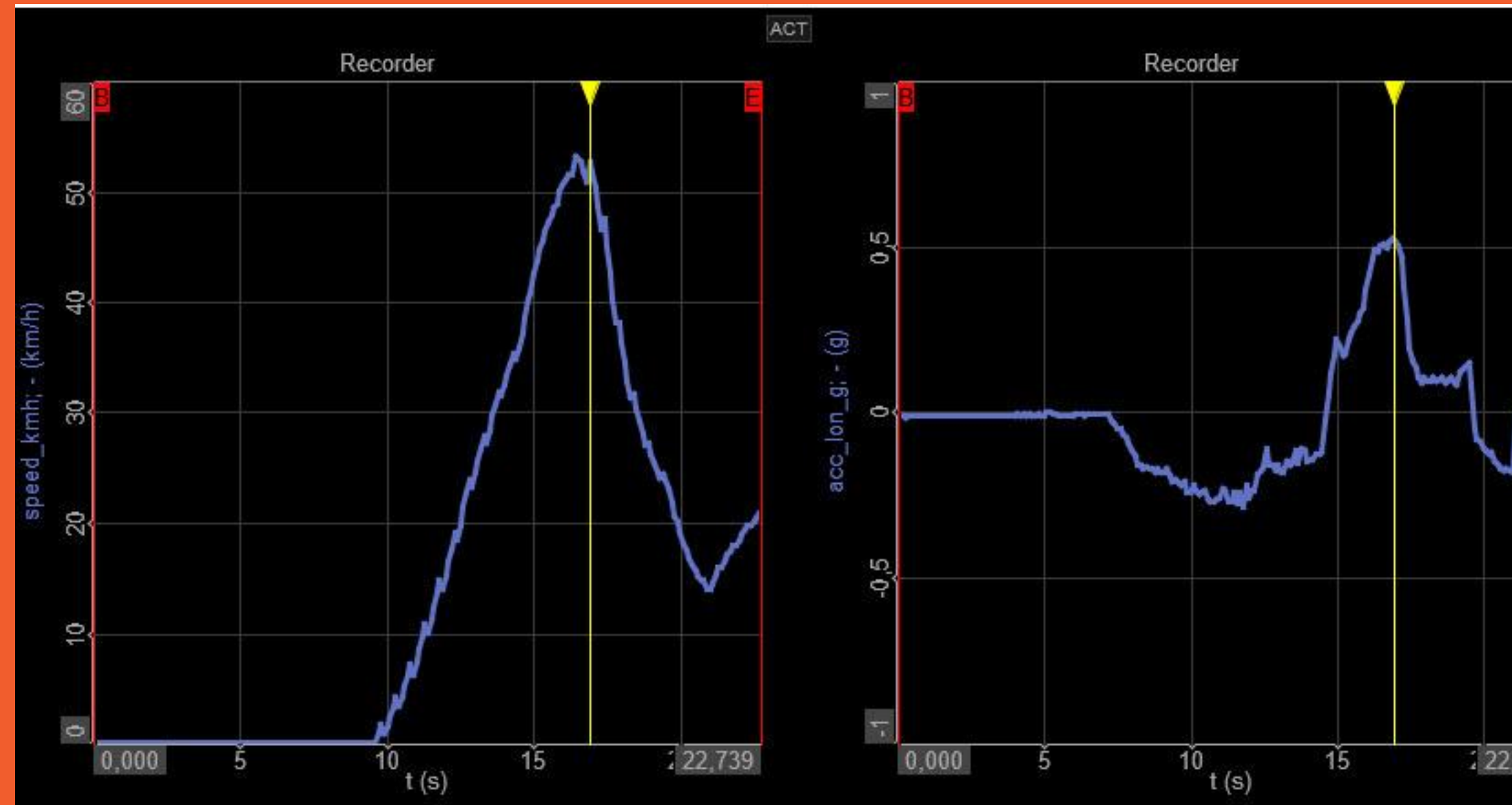
0 – 40 – 0

- Timer threshold value higher than noise: 0,5 km/h
- ABS disabled



! Physical data inconsistency

- Inconsistency between longitudinal acceleration and speed



cause: incorrect IMU mounting settings

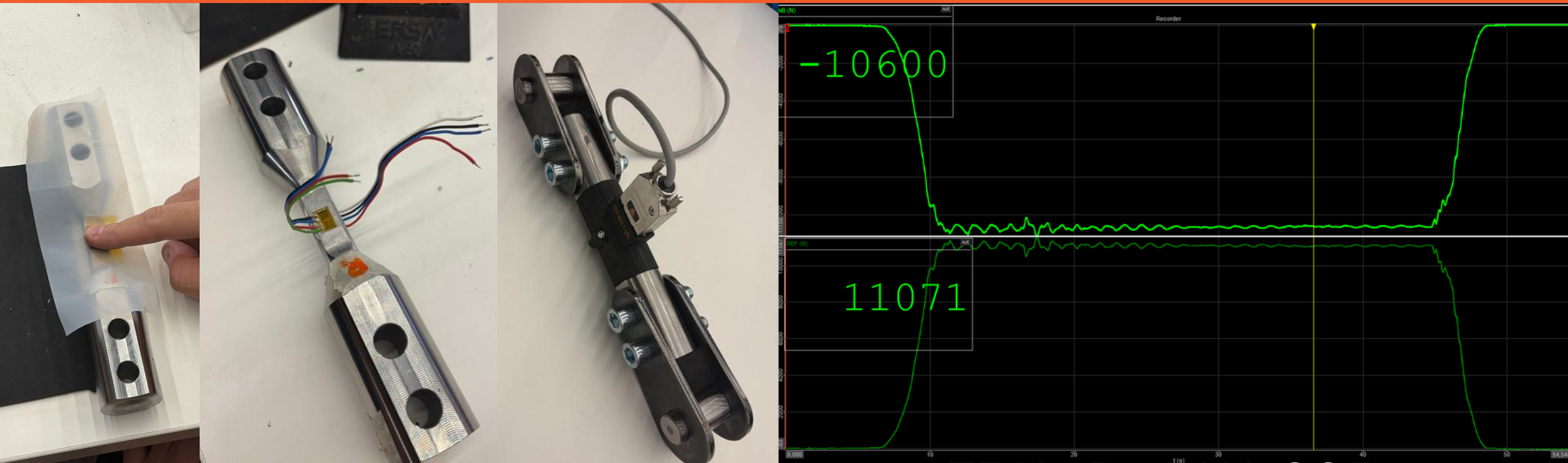
BUILD AN ACCURATE LOAD CELL

SOLUTION – PROGRESS – DIFFICULTIES – LEARNINGS

- Two un-wired T-rosette to create a full bridge
- Advanced load cell base
- Calculate sensitivity mathematically:

$$S = \frac{\frac{U_o}{U_e}}{F} = \frac{k_1 + k_3 + \nu(k_2 + k_4)}{(4 \cdot A \cdot E)}$$

- Accurate final results: 4% deviation
- The negative sign does not invalidate the measurement, as it is simply a matter of sign convention



STATIC SOUND MEASUREMENT

SOLUTION – PROGRESS – DIFFICULTIES – LEARNINGS

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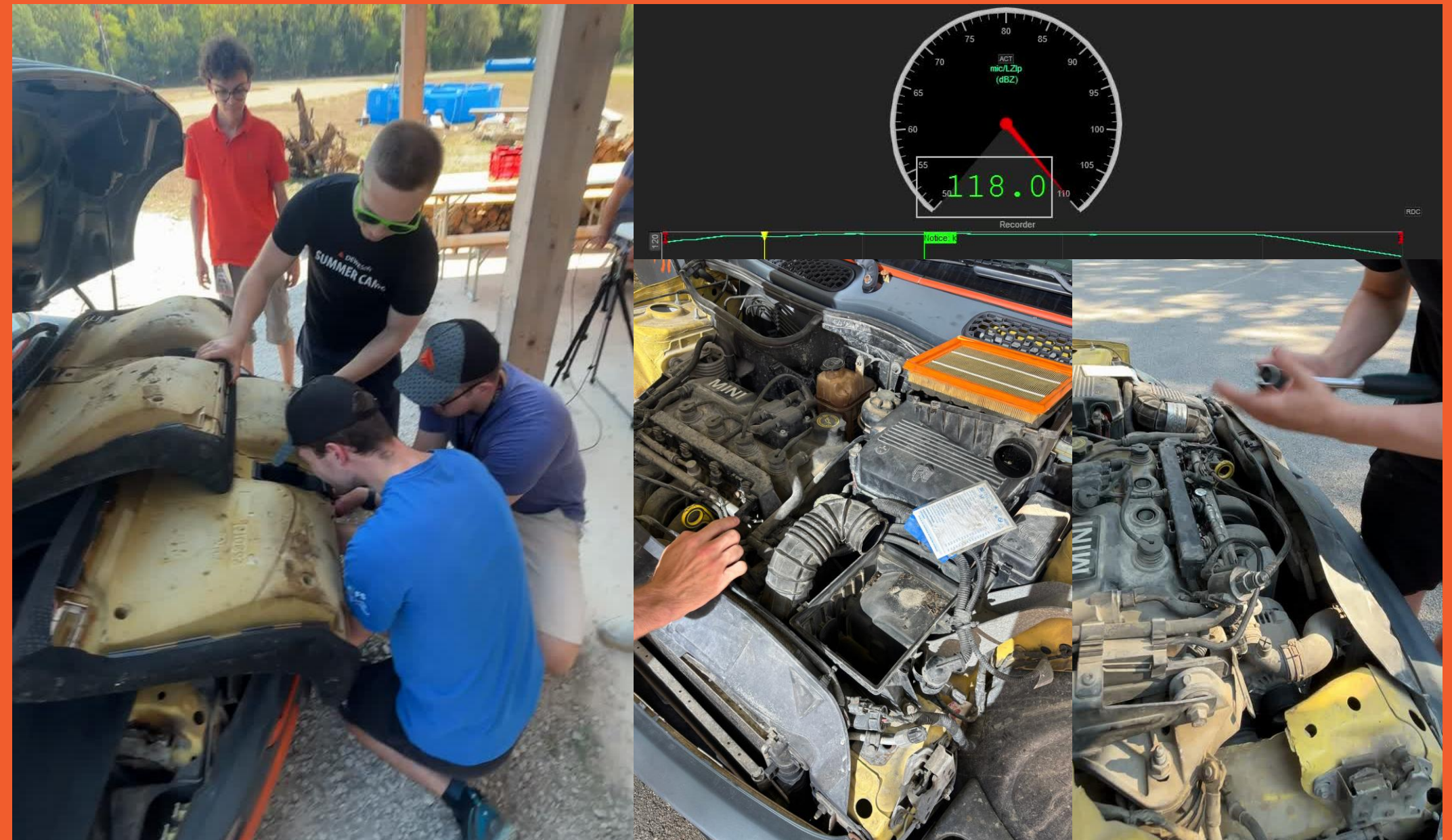


First measurement – LOUD:

- Removing two spark plugs
- Removing everything from intake to (including) the air filter
- Z frequency weighting
- Impulse time weighting

Secound measurement – QUIET:

- Covering the intake and the engine
- A frequency weighting
- Slow time weighting



TEAM-DEFINED CHALLENGE: G-FORCE PLOT

SOLUTION – PROGRESS – DIFFICULTIES – LEARNINGS

Utilising the DEWESoft SIRIUS DAQ to simultaneously measure:

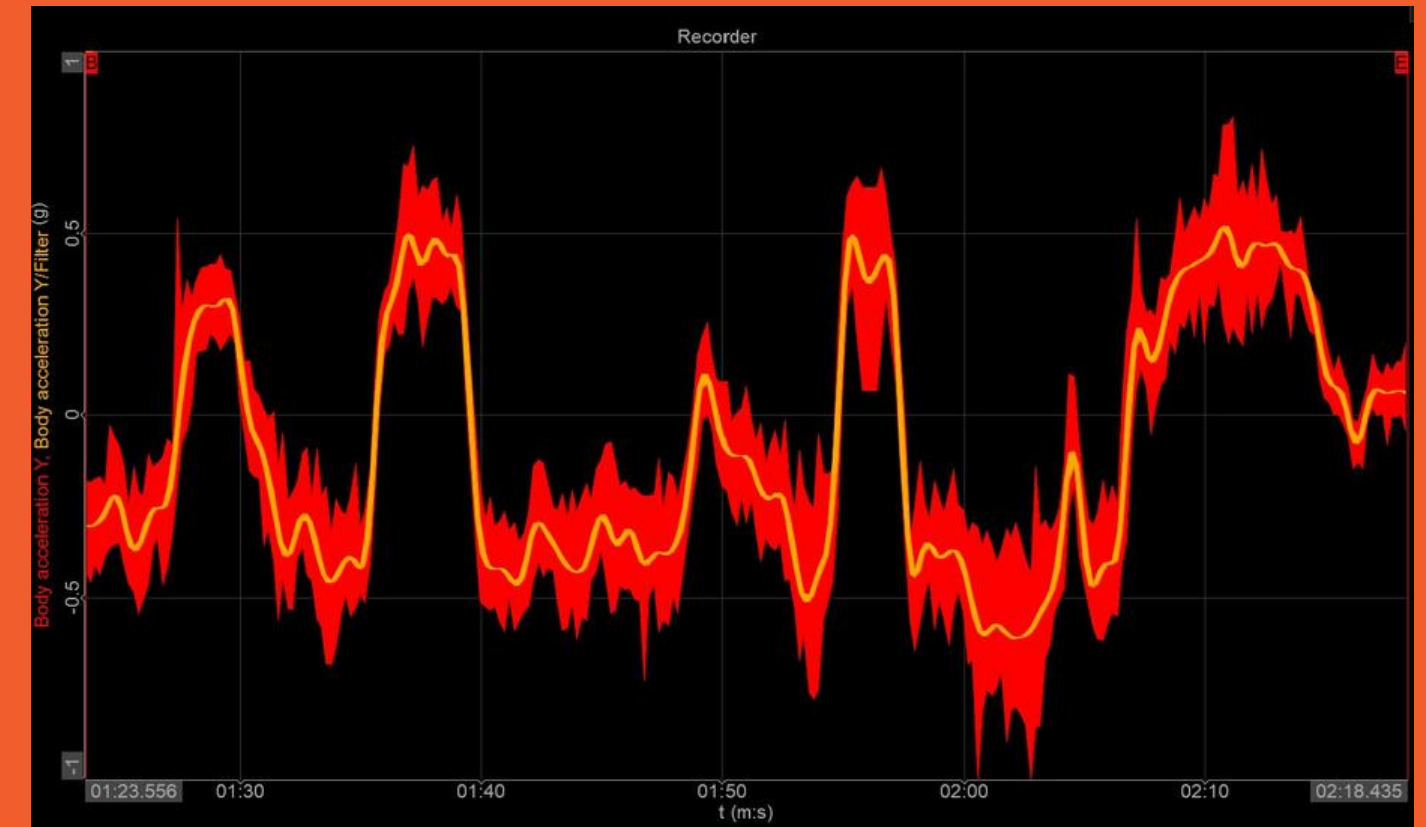
- Heading Angle
- Lateral G Force
- Wheels Speeds
- Steering Angle

Issues with steering angle data due to DEWESoftX bug (confirmed by Andy and Thorsten!)

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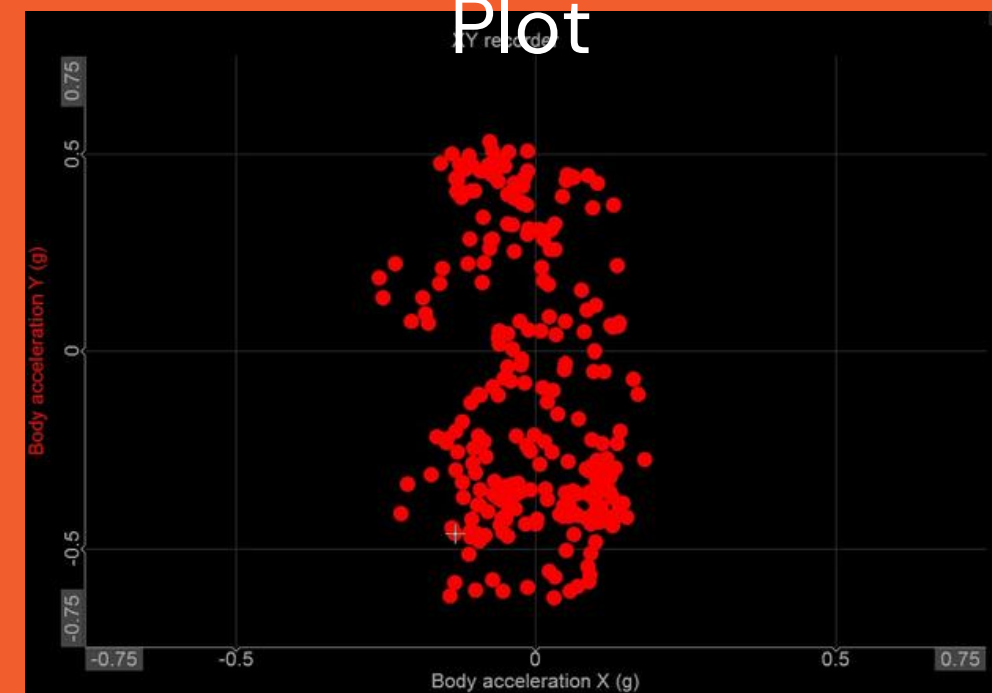


Lat G / Time



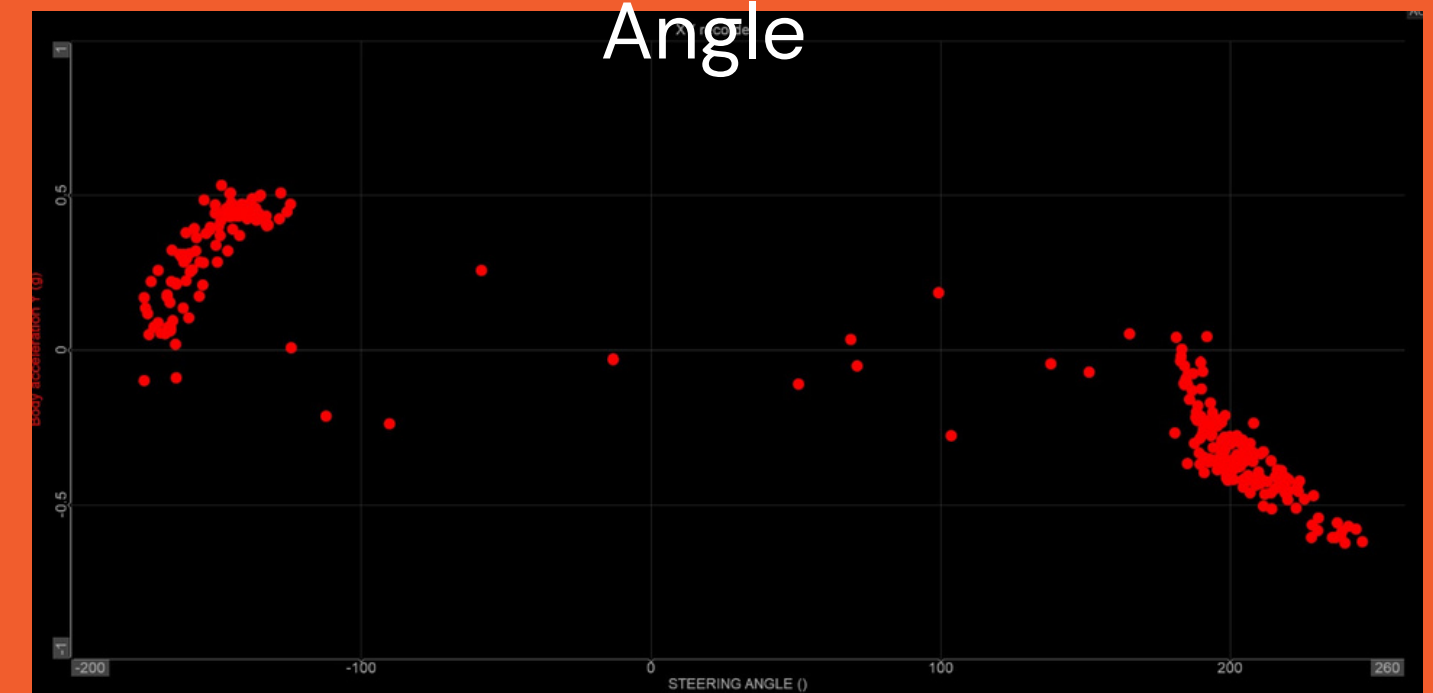
X Y G-G

Plot



Lat G / Steering

Angle



CAN

- Experimental validation of signals
- Logs for decoding
- Scaling and offsets from tests

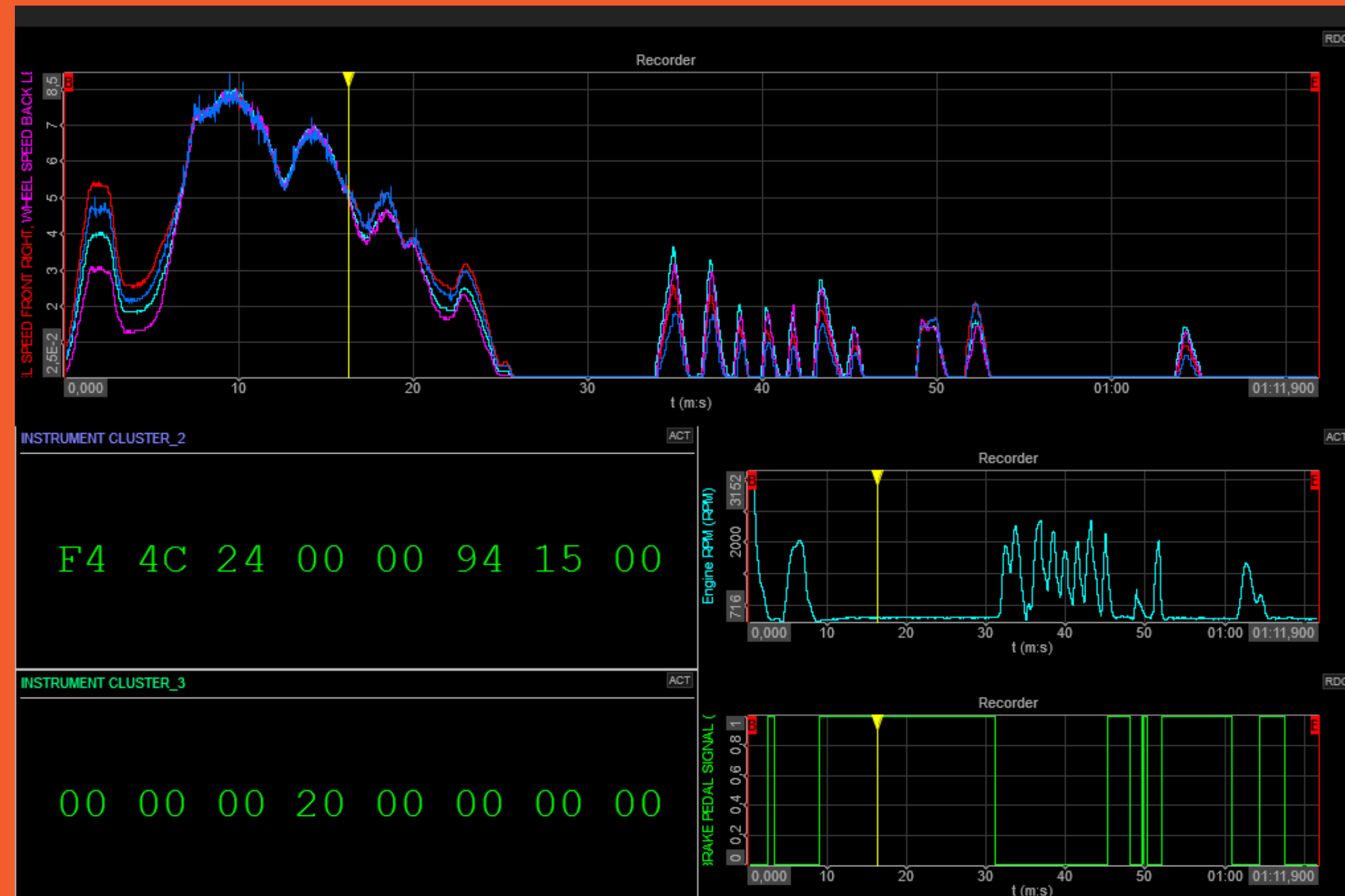


WHEEL SPEED FROM	7	6	5	4	3	2	1	0
COUNTER	0 7 0 6 0 5 0 4 0 3 0 2 0 1 0 lsb							
WHEEL SPEED FROM	0 15 0 14 0 lsb							
WHEEL SPEED BACK	0 23 0 22 0 21 0 20 0 19 0 18 0 17 0 lsb							
WHEEL SPEED BACK	0 31 0 30 0 29 0 28 0 27 0 26 0 25 0 24							
	0 39 0 38 0 37 0 36 0 35 0 34 0 33 0 lsb							
	0 47 0 46 0 45 0 44 0 43 0 42 0 41 0 40							
	0 55 0 54 0 53 0 52 0 51 0 50 0 49 0 lsb							
	0 63 0 62 0 61 0 60 0 59 0 58 0 57 0 56							

Add

Delete

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OTHER CHALLENGES

FUEL ESTIMATION



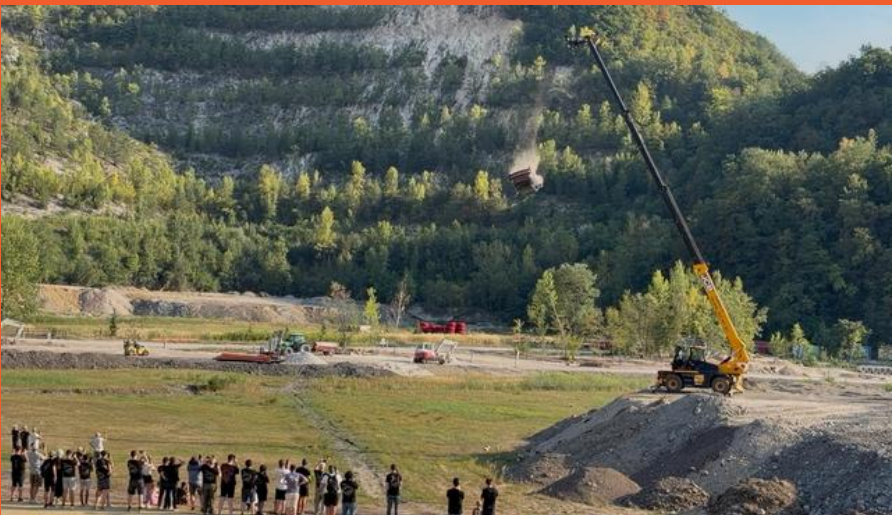
INSTAGRAM



FINANCES

Initial DSE Budget	10000
Total expenses	10600
Total revenues	3100
Net change	-7500
Current / Final DSE Balance	2500

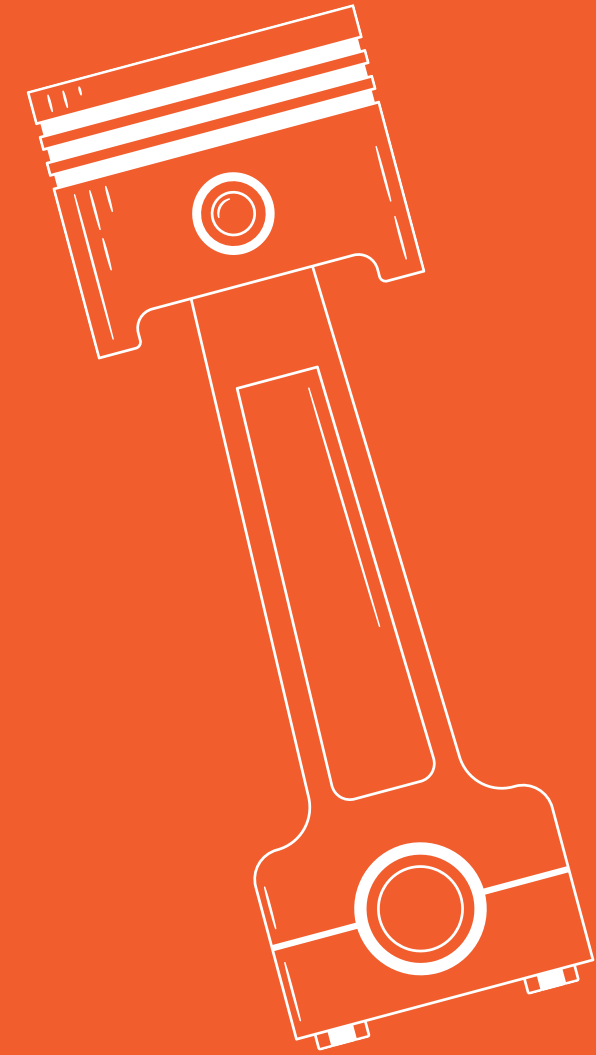
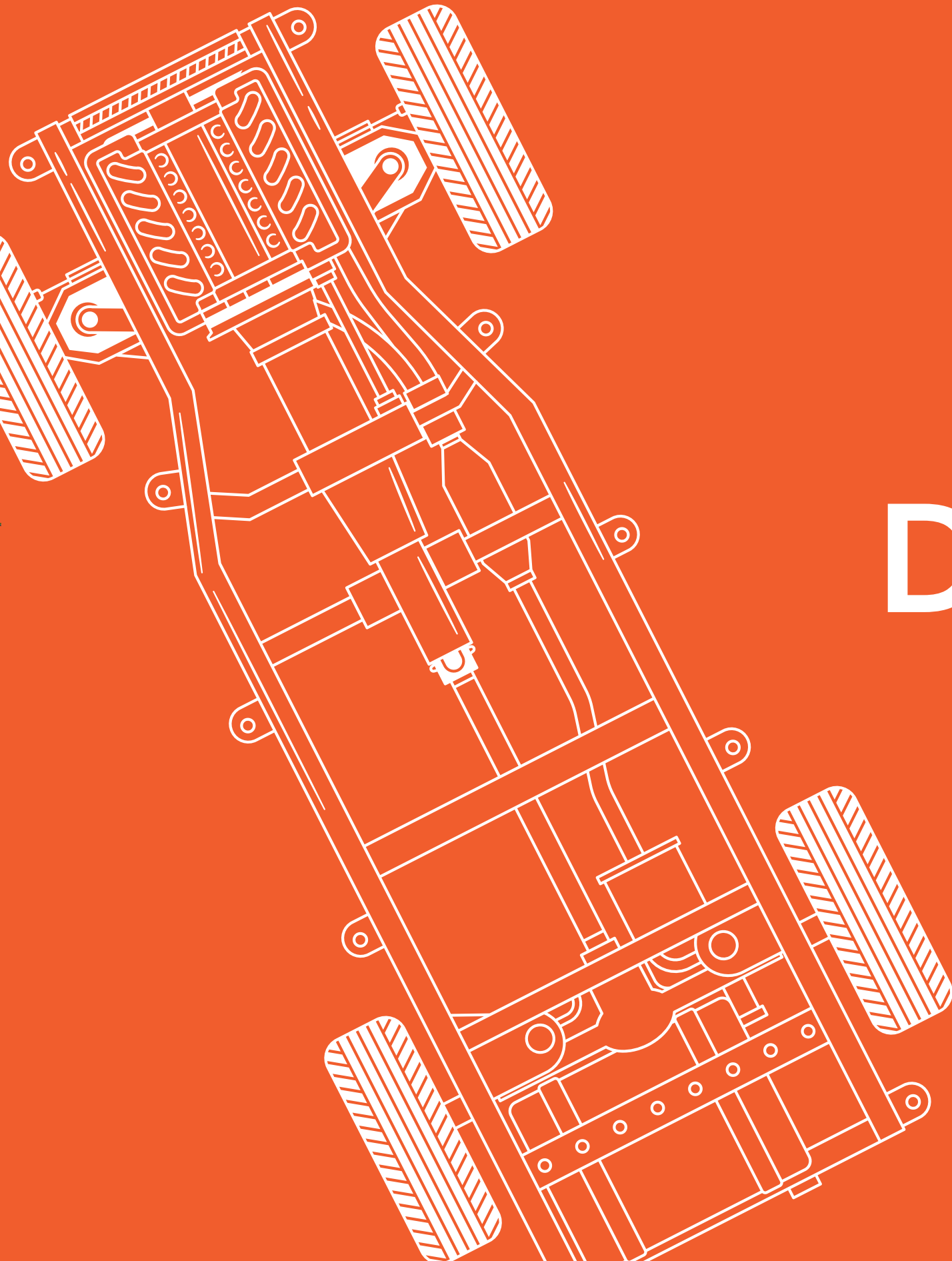
MARKETING CHALLENGE





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CONCLUSIONS





What we learnt as a Team:

- CAN decoding
- Wiring
- Basic Engine knowledge
- Application of strain gauges
- Creating setups in DEWESoftX
- Recording and analysing data in DEWESoft X
- Gravel driving skills
- Sound filters
- Set up of Sirius DAQ
- Set up and initialisation of IMU
- Use of Math function within DEWESoftX





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THANK YOU FOR
YOUR ATTENTION!

